

Characteristics of S-curve motion profile for all ranges of motion length and limits

Nguyen Quang Nam
Department of Mechanical Engineering
National University of Singapore
Singapore
quangnam.nguyen@u.nus.edu

Abstract—Motion control profiles (or trajectories) have been developed for decades. Among all types of motion profiles, trapezoidal profile, which is the fastest motion profile under velocity and acceleration limits, has been found to be inaccurate. The attention to jerk limit has led to the development of S-curve profile. Although many research works have been done to solve for the time-optimal S-curve profile for point-to-point motion, either analytically or computationally, the different shapes of it have not been deeply and clearly investigated. Therefore, the aim of this paper is to solve for the time-optimal S-curve profile analytically by categorizing it into four types based on its shapes and also analyze the characteristics of S-curve profile in terms of transfer time and electrical energy loss for different ranges of motion length and limits.

Keywords—motion control, S-curve profile, time-optimal motion profile planning, point-to-point motion, transfer time, energy loss.

I. INTRODUCTION

Motion of many parts in mechatronic systems such as robots, CNC machines, and pick-and-place machines is point-to-point motion. In point-to-point motion, the actuator moves from one position to another position in such a way that it starts from rest and completes a full stop at the end. The motion can be translational or rotational. Since a three-dimensional translational motion can be decomposed into three components in X, Y, and Z directions (similarly, a rotation can be decomposed into rotations about three axes), the study of one-dimensional translational point-to-point motion is sufficient for programming the three-dimensional motion of the machine. The translational quantities such as position, velocity, acceleration, and jerk (first-order derivative of acceleration) are analogous to angular displacement, angular velocity, angular acceleration, and angular jerk for rotational motion, e.g., motion of motors.

Motion profile is the set of equations of motion that describes the motion, including jerk, acceleration, velocity, and position as functions of time:

$$j = j(t), a = a(t), v = v(t), x = x(t) \quad (1)$$

with $t = 0$ when the system starts from rest at $x = 0$ and $t = T$ when it stops completely at the target position $x = L$. Here, the total time T needed to perform the motion is defined as “transfer time,” and the displacement L from the start position to the stop position is defined as “motion length.”

An actual servo (or motor) always has limits for velocity, acceleration, and jerk, denoted V_{max} , A_{max} , and J_{max} respectively, which means it can operate normally and safely when the maximum magnitudes of its velocity, acceleration, and jerk, denoted V , A , and J respectively, do not exceed the respective limits, i.e.,

$$J \leq J_{max}, A \leq A_{max}, V \leq V_{max} \quad (2)$$

A motion profile has one or some parameters in the equations of motion so that the system can perform a task in many different ways by changing the parameter(s). Motion profile planning is the method of tuning the parameters to achieve a purpose. The planning purpose of the time-optimal planning in many practical situations is to complete the point-to-point movement over a motion length L as fast as possible (i.e. to minimize transfer time T) while keeping the velocity, acceleration, and jerk under the respective limits as in (2).

In this paper, an analytical method of planning the time-optimal S-curve profile will be introduced. The characteristics of S-curve profile will be analyzed in terms of transfer time and electrical energy loss during the motion over different ranges of motion length for a given set of limits. Numerical plots of the results are obtained in MATLAB. Furthermore, effects of jerk limit and the connection between S-curve and trapezoidal curve will be discussed.

A. Trapezoidal profile

In the early years of motion control, jerk limit did not get much attention and only limits for velocity and acceleration were set for the machines. Under velocity limit V_{max} and acceleration limit A_{max} only, the trapezoidal profile, which uses maximum acceleration to achieve a maximum velocity and uses maximum deceleration to slow down to rest, produces the fastest motion. Indeed, planning time-optimal profile requires minimizing the transfer time under following constraints: the initial and final values of velocity must be zero, maximum height of the velocity curve is V_{max} , maximum magnitude of the slope of velocity curve is under A_{max} , and the area under velocity curve has to be equal to the motion length L . Finding the optimal solution of this constraint problem is analogous to maximizing the area under velocity curve such that: the two ends of the velocity curve are fixed at $(t = 0, v = 0)$ and $(t = T, v = 0)$, maximum height of the velocity curve is V_{max} , maximum magnitude of the slope of velocity curve is under A_{max} . Solution to this optimization problem, which is the fastest profile under velocity and acceleration limits, is the trapezoidal profile as shown in Fig. 1.

The time-optimal trapezoidal profile planning has been done in [1]. For long motion length, defined as $L \geq V_{max}^2/A_{max}$, both acceleration and velocity limits are reached and the total transfer time is $T = L/V_{max} + V_{max}/A_{max}$. The constant-velocity part of velocity curve will be eliminated in case of short motion length, i.e., $L < V_{max}^2/A_{max}$. Although acceleration is set at its limit A_{max} , the maximum velocity during the motion is still under the limit, i.e., $V = \sqrt{LA_{max}} < V_{max}$. The transfer time in this case is $T = 2\sqrt{L/A_{max}}$.

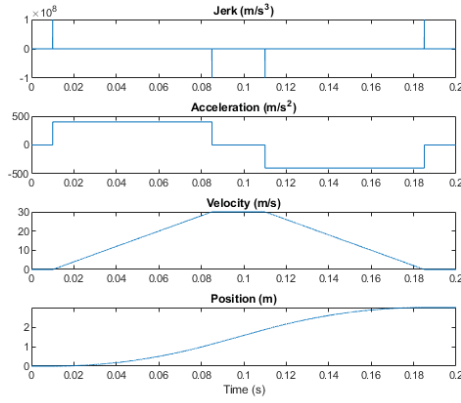


Fig. 1. Trapezoidal profile.

However, by differentiating velocity function twice, the obtained jerk function takes infinite value whenever the acceleration changes abruptly. In practical, an abrupt change of acceleration divided by small time step results in a very high value of jerk as in Fig. 1. Since acceleration is directly associated with forces or torques, many studies have shown that high jerks which corresponds to abrupt changes of acceleration tend to excite the system vibration, therefore cause overshoot [1], inaccuracy, and other harmful effects on the machine [2]. To avoid that, a limit for jerk needs to be added into the set of velocity and acceleration limits [1].

B. S-curve profile

Although trapezoidal profile is the fastest profile under only acceleration and velocity limits, it is not accurate and reliable due to abrupt changes of acceleration [3]. The attempts to avoid that by introducing jerk limit had led to the development of S-curve profile (also known as double-S profile). S-curve motion profile is the modification of trapezoidal profile which replaces each abrupt increase and decrease of acceleration by a ramp-up and ramp-down, respectively. The slopes of these ramp-ups and ramp-downs correspond to constant jerk values. A general S-curve profile consists of 7 segments as shown in Fig. 2.

The ramp-up/ramp-down time of acceleration is called “constant-jerk time” T_J , and obviously $T_J > 0$ because if not, the profile becomes trapezoidal profile. The time during which

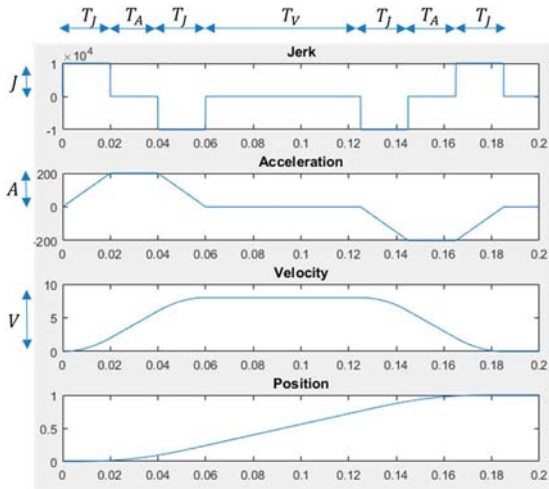


Fig. 2. General form of S-curve profile consists of 7 segments.

acceleration remains unchanged is called “constant-acceleration time” T_A . The time interval during which velocity remains unchanged is called “constant-velocity time” T_V . The transfer time is related to these time intervals as

$$T = 4 T_J + 2 T_A + T_V \quad (3)$$

II. PLANNING THE TIME-OPTIMAL S-CURVE PROFILE

Many studies have been done on planning the S-curve profile for different purposes. In [4], an algorithm is used to optimize the S-curve profile for better energy consumption using a mass-spring-damper model. A modified S-curve profile using sinusoidal jerk [5] and an nth-order S-curve [6] are proposed to reduce residual vibration. Many other research works focus on planning the time-optimal S-curve profile, e.g., using computer algorithms to optimize S-curve with trigonometric jerk model [7] or sigmoid jerk model [8], using control loop to constrains the limits in real time [9]. Computer algorithms can also be used to plan asymmetric S-curve [10] or a general motion profile [11]. Although these computational methods are more capable of solving complex problems, planning the time-optimal S-curve can be solved using analytical method which gives the exact solutions and reveals some characteristics of S-curve profile.

The optimization problem to be solved is: find the optimal values of the parameters V, A, T_V, T_A, T_J of the equations of motion in terms of: the chosen jerk value J , the limits V_{max}, A_{max} , and the required motion length L such that the transfer time T is minimized.

A. General equations of motion

The equations of motion of the general S-curve profile must be expressed in terms of the maximum magnitudes V, A, J and the time intervals T_V, T_A, T_J . The general S-curve profile consists of 7 segments. Firstly, the function of jerk is

$$j(t) = \begin{cases} J, & 0 < t \leq T_1 \\ 0, & T_1 < t \leq T_2 \\ -J, & T_2 < t \leq T_3 \\ 0, & T_3 < t \leq T_4 \\ -J, & T_4 < t \leq T_5 \\ 0, & T_5 < t \leq T_6 \\ J, & T_6 < t \leq T_7 \end{cases} \quad (4)$$

where

$$\begin{cases} T_1 = T_J \\ T_2 = T_J + T_A \\ T_3 = 2T_J + T_A \\ T_4 = 2T_J + T_A + T_V \\ T_5 = T - T_J - T_A \\ T_6 = T - T_J \\ T_7 = T \end{cases} \quad (5)$$

For convenience, define

$$\Delta t_i = t - T_{i-1} \text{ for } T_{i-1} < t \leq T_i \quad (6)$$

the relative time with respect to the beginning of each segment.

Integrating the above jerk equations with respect to Δt_i from 0 to $(T_i - T_{i-1})$ for each segment (follow the sequence from segment 1 to segment 7), the acceleration equations are obtained as

$$a(t) = \begin{cases} J\Delta t_1, & 0 < t \leq T_1 \\ A, & T_1 < t \leq T_2 \\ A - J\Delta t_3, & T_2 < t \leq T_3 \\ 0, & T_3 < t \leq T_4 \\ -J\Delta t_5, & T_4 < t \leq T_5 \\ -A, & T_5 < t \leq T_6 \\ -A + J\Delta t_7, & T_6 < t \leq T_7 \end{cases} \quad (7)$$

where the maximum magnitude of acceleration is obtained by substituting $\Delta t_1 = T_j$ into the first equation in (7)

$$A = JT_j \quad (8)$$

Similarly, integrating the above acceleration equations with respect to Δt_i from 0 to $(T_i - T_{i-1})$ for each segment (follow the sequence from segment 1 to segment 7), the velocity equations are:

$$v(t) = \begin{cases} \frac{1}{2}J\Delta t_1^2, & 0 < t \leq T_1 \\ \frac{1}{2}AT_j + A\Delta t_2, & T_1 < t \leq T_2 \\ A\left(\frac{T_j}{2} + T_A\right) + A\Delta t_3 - \frac{1}{2}J\Delta t_3^2, & T_2 < t \leq T_3 \\ V, & T_3 < t \leq T_4 \\ V - \frac{1}{2}J\Delta t_5^2, & T_4 < t \leq T_5 \\ V - \frac{1}{2}AT_j - A\Delta t_6, & T_5 < t \leq T_6 \\ \frac{1}{2}AT_j - A\Delta t_7 + \frac{1}{2}J\Delta t_7^2, & T_6 < t \leq T_7 \end{cases} \quad (9)$$

The maximum velocity is obtained by substituting $\Delta t_3 = T_j$ into the third equation in (9)

$$V = A(T_j + T_A) \quad (10)$$

Finally, integrating the above velocity equations with respect to Δt_i from 0 to $(T_i - T_{i-1})$ for each segment (follow the sequence from segment 1 to segment 7), the position equations are obtained as

$x(t) =$

$$\begin{cases} \frac{1}{6}J\Delta t_1^3, & 0 < t \leq T_1 \\ \frac{1}{6}AT_j^2 + \frac{1}{2}AT_j\Delta t_2 + \frac{1}{2}A\Delta t_2^2, & T_1 < t \leq T_2 \\ \frac{1}{6}AT_j^2 + \frac{1}{2}VT_A + \left(V - \frac{1}{2}AT_j\right)\Delta t_3 + \frac{1}{2}A\Delta t_3^2 - \frac{1}{6}J\Delta t_3^3, & T_2 < t \leq T_3 \\ V\left(T_j + \frac{T_A}{2}\right) + V\Delta t_4, & T_3 < t \leq T_4 \\ V\left(T_j + \frac{T_A}{2}\right) + VT_V + V\Delta t_5 - \frac{1}{6}J\Delta t_5^3, & T_4 < t \leq T_5 \\ V\left(2T_j + \frac{T_A}{2} + T_V\right) - \frac{1}{6}AT_j^2 + \left(V - \frac{1}{2}AT_j\right)\Delta t_6 - \frac{1}{2}A\Delta t_6^2, & T_5 < t \leq T_6 \\ V(2T_j + T_A + T_V) - \frac{1}{6}AT_j^2 + \frac{1}{2}AT_j\Delta t_7 - \frac{1}{2}A\Delta t_7^2 + \frac{1}{6}J\Delta t_7^3, & T_6 < t \leq T_7 \end{cases} \quad (11)$$

The motion length can be obtained by substituting $\Delta t_7 = T_j$ into the last equation in (11)

$$L = V(2T_j + T_A + T_V) \quad (12)$$

Solving (8), (10), and (12) for T_j, T_A, T_V in terms of V, A, J

$$T_j = \frac{A}{J}, \quad T_A = \frac{V}{A} - \frac{A}{J}, \quad T_V = \frac{L}{V} - \frac{V}{A} - \frac{A}{J} \quad (13)$$

Substitute (13) into (3)

$$T = \frac{L}{V} + \frac{V}{A} + \frac{A}{J} \quad (14)$$

Substitute (8) and (10) into (12)

$$L = JT_j(T_j + T_A)(2T_j + T_A + T_V) \quad (15)$$

The relations in (10), (13), (14), and (15) are the fundamental relations in the S-curve profile and will be used to determine the parameters for 4 types of S-curve in section II.C.

B. Four types of S-curve for time-optimal motion profile

Since the magnitude of jerk is fixed, the time-optimal profile is categorized into four types depending on whether velocity and acceleration limits are reached:

1) *The 7-segment S-curve profile*: Both acceleration and velocity limits are reached ($A = A_{max}$, $V = V_{max}$) which means constant-acceleration and constant-velocity time are not eliminated ($T_A > 0$, $T_V > 0$). In this case, the profile still consists of 7 segments as shown in Fig. 3.

2) *The 5-segment type-A S-curve profile*: Only acceleration limit is reached ($A = A_{max}$, $T_A > 0$) while velocity limit is not reached ($V \leq V_{max}$, $T_V = 0$). In this case, the profile consists of 5 segments as shown in Fig. 4.

3) *The 5-segment type-B S-curve profile*: Only velocity limit is reached ($V = V_{max}$, $T_V > 0$) while acceleration limit is not reached ($A \leq A_{max}$, $T_A = 0$). In this case, the profile also consists of 5 segments as shown in Fig. 5.

4) *The 3-segment S-curve profile*: Both constant-acceleration and constant-velocity time are eliminated ($T_A = 0$, $T_V = 0$) while acceleration and velocity limits are not reached ($A \leq A_{max}$, $V \leq V_{max}$). In this case, the profile consists of 3 segments as shown in Fig. 6.

C. Determine the parameters for 4 types of S-curve

This section aims to find all the parameters V, A, T_V, T_A, T_j in terms of the chosen jerk magnitude J , the given limits V_{max}, A_{max} , and the motion length L for each type of S-curve.

1) *For the 7-segment S-curve profile*: Substitute $A = A_{max}$, $V = V_{max}$ into (13) and (14)

$$T_j = \frac{A_{max}}{J}, \quad T_A = \frac{V_{max}}{A_{max}} - \frac{A_{max}}{J}, \quad T_V = \frac{L}{V_{max}} - \frac{V_{max}}{A_{max}} - \frac{A_{max}}{J} \quad (16)$$

$$T = \frac{L}{V_{max}} + \frac{V_{max}}{A_{max}} + \frac{A_{max}}{J} \quad (17)$$

2) *For the 5-segment type-A S-curve profile*: Substitute $A = A_{max}$ into the first relation in (13)

$$T_j = \frac{A_{max}}{J} \quad (18)$$

Substitute (18) together with $A = A_{max}$, $T_V = 0$ into (15) and solve for T_A

$$T_A = \frac{1}{2} \sqrt{\left(\frac{A_{max}}{J}\right)^2 + \frac{4L}{A_{max}}} - \frac{3}{2} \frac{A_{max}}{J} \quad (19)$$

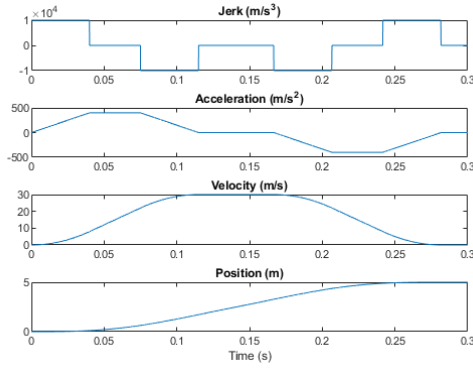


Fig. 3. The 7-segment S-curve profile, given $L = 5m$, $V_{max} = 30ms^{-1}$, $A_{max} = 400ms^{-2}$, $J_{max} = 10^4ms^{-3}$.

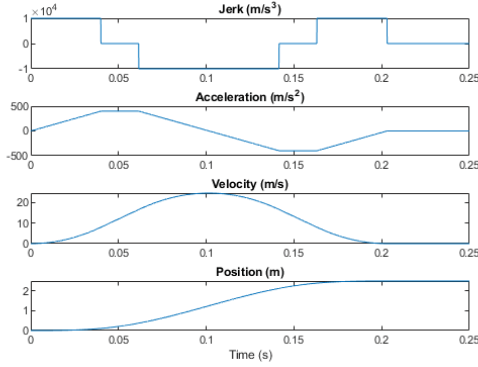


Fig. 4. The 5-segment type-A S-curve profile, given $L = 2.5m$, $V_{max} = 30ms^{-1}$, $A_{max} = 400ms^{-2}$, $J_{max} = 10^4ms^{-3}$.

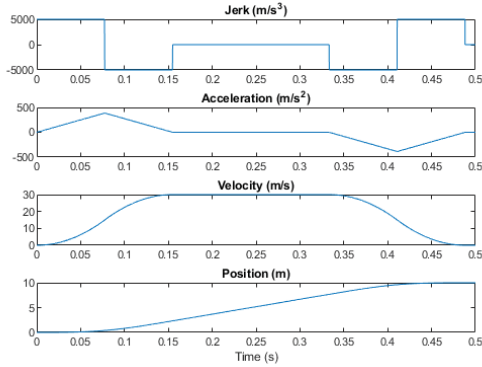


Fig. 5. The 5-segment type-B S-curve profile, given $L = 10m$, $V_{max} = 30ms^{-1}$, $A_{max} = 500ms^{-2}$, $J_{max} = 5 \times 10^3ms^{-3}$.

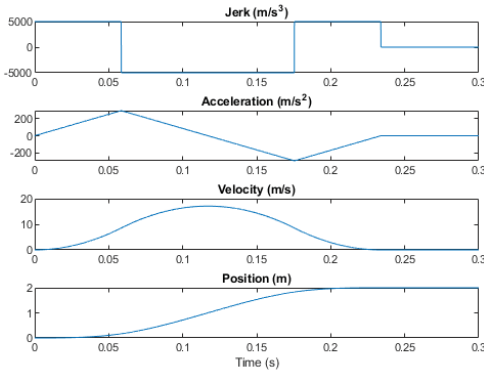


Fig. 6. The 3-segment S-curve profile, given $L = 2m$, $V_{max} = 30ms^{-1}$, $A_{max} = 400ms^{-2}$, $J_{max} = 5 \times 10^3ms^{-3}$.

Finally, substitute (18) and (19) into (10) and (14)

$$V = \frac{A_{max}}{2} \left(\sqrt{\left(\frac{A_{max}}{J} \right)^2 + \frac{4L}{A_{max}}} - \frac{A_{max}}{J} \right) \quad (20)$$

$$T = \sqrt{\left(\frac{A_{max}}{J} \right)^2 + \frac{4L}{A_{max}}} + \frac{A_{max}}{J} \quad (21)$$

3) For the 5-segment type-B S-curve profile: Substitute $V = V_{max}$, $T_A = 0$ into the second relation in (13)

$$A = \sqrt{V_{max} J} \quad (22)$$

Then substitute (22) into the other relations in (13)

$$T_J = \frac{\sqrt{V_{max}}}{J}, \quad T_V = \frac{L}{V_{max}} - 2\sqrt{\frac{V_{max}}{J}} \quad (23)$$

Finally, substitute (22) and (23) into (14)

$$T = \frac{L}{V_{max}} + 2\sqrt{\frac{V_{max}}{J}} \quad (24)$$

4) For the 3-segment S-curve profile: Substitute $T_A = 0$, $T_V = 0$ into (15)

$$T_J = \sqrt[3]{\frac{L}{2J}} \quad (25)$$

Then substitute (25) into the first relation in (13)

$$A = \sqrt[3]{\frac{LJ^2}{2}} \quad (26)$$

Finally, substitute (25) and (26) into (10) and (14)

$$V = \sqrt[3]{\frac{L^2 J}{4}}, \quad T = \sqrt[3]{\frac{32L}{J}} \quad (27)$$

Based on the formulas for transfer time in (17), (21), (24), and (27), it is obvious that the transfer time decreases as the magnitude of jerk increases. Hence, for the time-optimal profile, choose

$$J = J_{max} \quad (28)$$

Summary of the formulas for calculating parameters for 4 types of S-curve is shown in Table I.

D. Optimal S-curve for different motion lengths and limits

This section makes use of the results in section II.B and II.C to determine which type of S-curve should be used for different values of $L, V_{max}, A_{max}, J_{max}$.

Define the following lengths and ratio:

$$L_1 = \frac{2A_{max}^3}{J_{max}^2}, \quad L_2 = 2\sqrt{\frac{V_{max}^3}{J_{max}}}, \quad L_3 = \frac{V_{max}^2}{A_{max}} + \frac{V_{max}A_{max}}{J_{max}} \quad (29)$$

$$R = \frac{V_{max} J_{max}}{A_{max}^2} \quad (30)$$

TABLE I. PARAMETERS CALCULATED IN TERMS OF THE GIVEN DATA $L, V_{max}, A_{max}, J_{max}$ FOR 4 TYPES OF S-CURVE.

	3 segments	5 segments A	5 segments B	7 segments
J	J_{max}			
A	$\sqrt[3]{\frac{L J_{max}^2}{2}}$	A_{max}	$\sqrt{V_{max} J_{max}}$	A_{max}
V	$\sqrt[3]{\frac{L^2 J_{max}}{4}}$	$\frac{A_{max}}{2} \left(\sqrt{\left(\frac{A_{max}}{J_{max}}\right)^2 + \frac{4L}{A_{max}}} - \frac{A_{max}}{J_{max}} \right)$	V_{max}	V_{max}
T_J	$\sqrt[3]{\frac{L}{2 J_{max}}}$	$\frac{A_{max}}{J_{max}}$	$\sqrt{\frac{V_{max}}{J_{max}}}$	$\frac{A_{max}}{J_{max}}$
T_A	0	$\frac{1}{2} \sqrt{\left(\frac{A_{max}}{J_{max}}\right)^2 + \frac{4L}{A_{max}}} - \frac{3 A_{max}}{2 J_{max}}$	0	$\frac{V_{max}}{A_{max}} - \frac{A_{max}}{J_{max}}$
T_V	0	0	$\frac{L}{V_{max}} - 2 \sqrt{\frac{V_{max}}{J_{max}}}$	$\frac{L}{V_{max}} - \frac{V_{max}}{A_{max}} - \frac{A_{max}}{J_{max}}$
T	$\sqrt[3]{\frac{32L}{J_{max}}}$	$\sqrt{\left(\frac{A_{max}}{J_{max}}\right)^2 + \frac{4L}{A_{max}}} + \frac{A_{max}}{J_{max}}$	$\frac{L}{V_{max}} + 2 \sqrt{\frac{V_{max}}{J_{max}}}$	$\frac{L}{V_{max}} + \frac{V_{max}}{A_{max}} + \frac{A_{max}}{J_{max}}$

1) For the 7-segment S-curve profile: From section II.B, the conditions are

$$\begin{cases} A \leq A_{max} \\ V \leq V_{max} \end{cases} \quad (37)$$

$$\begin{cases} T_A > 0 \\ T_V > 0 \end{cases} \quad (31)$$

Substitute the formulas in Table I into (37)

Substitute the formulas in Table I into (31)

$$\begin{cases} \frac{V_{max} J_{max}}{A_{max}^2} > 1 \\ L > \frac{V_{max}^2}{A_{max}} + \frac{A_{max} V_{max}}{J_{max}} \end{cases} \Rightarrow \begin{cases} R > 1 \\ L > L_3 \end{cases} \quad (32)$$

$$\begin{cases} L \leq \frac{2A_{max}^3}{J_{max}^2} \\ L \leq 2 \sqrt{\frac{V_{max}^3}{J_{max}}} \end{cases} \Rightarrow \begin{cases} L \leq L_1 \\ L \leq L_2 \end{cases} \quad (38)$$

2) For the 5-segment type-A S-curve profile: From section II.B, the conditions are

$$\begin{cases} V \leq V_{max} \\ T_A > 0 \end{cases} \quad (33)$$

Substitute the formulas in Table I into (33)

$$\begin{cases} L \leq \frac{V_{max}^2}{A_{max}} + \frac{A_{max} V_{max}}{J_{max}} \\ L > \frac{2A_{max}^3}{J_{max}^2} \end{cases} \Rightarrow \begin{cases} L \leq L_3 \\ L > L_1 \end{cases} \quad (34)$$

3) For the 5-segment type-B S-curve profile: From section II.B, the conditions are

$$\begin{cases} A \leq A_{max} \\ T_V > 0 \end{cases} \quad (35)$$

Substitute the formulas in Table I into (35)

$$\begin{cases} \frac{V_{max} J_{max}}{A_{max}^2} \leq 1 \\ L > 2 \sqrt{\frac{V_{max}^3}{J_{max}}} \end{cases} \Rightarrow \begin{cases} R \leq 1 \\ L > L_2 \end{cases} \quad (36)$$

4) For the 3-segment S-curve profile: From section II.B, the conditions are

All conditions in (32), (34), (36), and (38) are used to determine the optimal S-curve as shown in Table II. Note that from (29) and (30): $L_2 \leq L_3 \leq L_1$ for $R \leq 1$ and $L_1 < L_2 < L_3$ for $R > 1$. The formulas in Table I and II are verified by the results obtained by using MATLAB Global Optimization Toolbox [12].

E. Steps on planning the time-optimal S-curve profile

The method of planning the time-optimal S-curve profile can be summarized in 3 steps below:

- From the given values $L, V_{max}, A_{max}, J_{max}$, calculate L_1, L_2, L_3, R using (29) and (30).
- From the values of R and the relation between L, L_1, L_2, L_3 , determine the suitable S-curve type using Table II.
- According to the S-curve type determined previously, calculate all parameters using Table I.

With all the values of V, A, J, T_V, T_A, T_J calculated in 3 steps above, the equations of motion in (4), (7), (9), and (11) are determined and can be used to program the motion of the machines. MATLAB can be used to plot the results for different sets of $L, V_{max}, A_{max}, J_{max}$ as in Fig. 3, 4, 5, and 6.

TABLE II. OPTIMAL S-CURVE TYPE FOR ALL POSSIBLE CASES.

$R \leq 1$		$R > 1$	
Motion length	S-curve type	Motion length	S-curve type
$L \leq L_2$	3 segments	$L \leq L_1$	3 segments
$L > L_2$	5 segments B	$L_1 < L \leq L_3$	5 segments A
		$L > L_3$	7 segments

III. CHARACTERISTICS OF S-CURVE PROFILE

A. Transfer time for all ranges of motion length and limits

If velocity, acceleration, and jerk limits are fixed, Table II shows that different motion lengths require different S-curve types for time-optimal motion. The corresponding transfer time can be determined using Table I.

For $R \leq 1$, the time-optimal S-curve profile is 3-segment for $L \leq L_2$ and 5-segment type-B for $L > L_2$, as in Fig. 7.

For $R > 1$, the time-optimal S-curve profile is 3-segment for $L \leq L_1$, 5-segment type-A for $L_1 < L \leq L_3$, and 7-segment for $L > L_3$, as in Fig. 8.

For fixed values of V_{max} , A_{max} , increasing J_{max} leads to an increase of R . Table I shows that the transfer time decreases as jerk limit increases, as in Fig. 9.

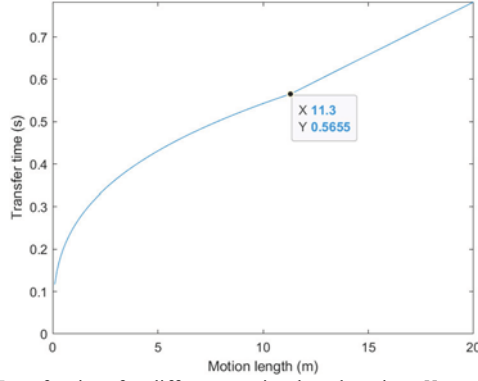


Fig. 7. Transfer time for different motion lengths, given $V_{max} = 40ms^{-1}$, $A_{max} = 400ms^{-2}$, $J_{max} = 2000ms^{-3}$, hence $R = 0.5$ and $L_2 = 11.3m$.

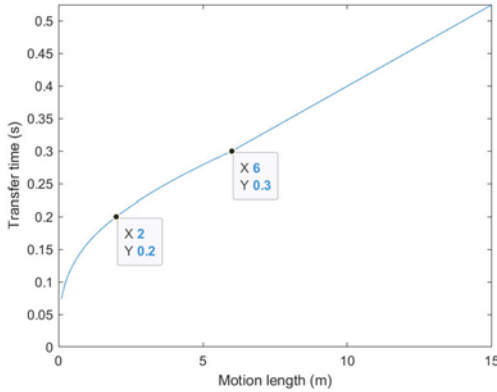


Fig. 8. Transfer time for different motion lengths, given $V_{max} = 40ms^{-1}$, $A_{max} = 400ms^{-2}$, $J_{max} = 8000ms^{-3}$, hence $R = 2$, $L_1 = 2m$, $L_3 = 6m$.

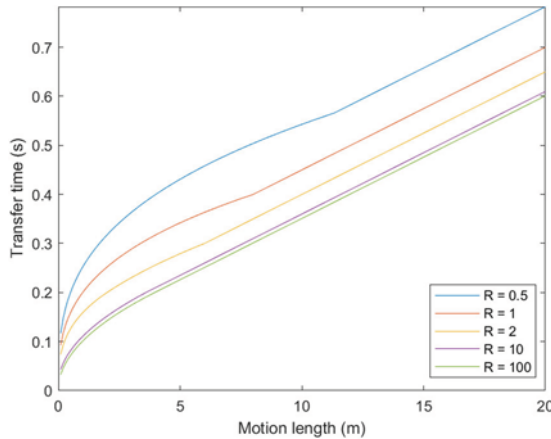


Fig. 9. Transfer time for different motion lengths for $R = 0.5, 1, 2, 10, 100$ by increasing J_{max} , given $V_{max} = 40ms^{-1}$, $A_{max} = 400ms^{-2}$.

B. S-curve profile with high jerk limit is trapezoidal profile

If the jerk limit is very high, i.e., $J_{max} \gg A_{max}^2/V_{max}$ or $R \gg 1$, then $L_1 \rightarrow 0$, $L_2 \rightarrow 0$, and $L_3 \rightarrow V_{max}^2/A_{max}$. Table II reduces to: the optimal S-curve is 5-segment type-A for $L \leq V_{max}^2/A_{max}$ and 7-segment for $L > V_{max}^2/A_{max}$. Table I shows that $T = 2\sqrt{L/A_{max}}$, $T_j = 0$ for 5-segment type-A S-curve and $T = L/V_{max} + V_{max}/A_{max}$, $T_j = 0$ for 7-segment S-curve. This is the same as trapezoidal profile. Since transfer time decreases as jerk limit increases as in Fig. 9, it shows that trapezoidal profile, which is the S-curve with infinite jerk limit, is the faster than S-curve profile.

C. Electric loss for all ranges of motion length and limits

Total energy consumption is the sum of mechanical work done to move a system's part over a motion length L and the electric loss due to electrical resistance. The mechanical work done is the sum of, firstly, the useful work done to accelerate/decelerate the mass m of the moving part:

$$W_a = \int_0^L (ma) dx = m \int_0^T a(vdt) = m \int_{v=0}^{v=0} v dv = 0 \quad (39)$$

and secondly, the work done by an approximately constant torque or force $\tau_c \propto F_c = \text{const}$ of the system to balance gravitational force, friction, and other external forces:

$$W_c = \int_0^L F_c dx = F_c L \quad (40)$$

Combining (39) and (40) shows that the mechanical work done does not depend on the motion profile. Therefore, the efficiency of the system performing point-to-point motion is only affected by the electric loss which is given by

$$E_{loss} = \int_0^T i^2(t) r dt \quad (41)$$

where $i(t)$ is the current in the circuit, $r = \text{const}$ is the generalized electrical resistance of the circuit [13]. Here, the current is proportional to the total torque or force generated, neglecting damping force:

$$i(t) \propto \tau_{total}(t) \propto F_{total}(t) = ma(t) + F_c \quad (42)$$

Substitute (42) into (41)

$$E_{loss} \propto m^2 \int_0^T a^2 dt + F_c^2 \int_0^T dt + 2mF_c \int_0^T a dt \quad (43)$$

Notice that $\int_0^T a dt = \int_0^T (dv/dt) dt = \int_{v=0}^{v=0} dv = 0$ then (43) reduces to

$$E_{loss} \propto m^2 \int_0^T a^2 dt + F_c^2 T \quad (44)$$

The second term in (44) show that minimizing transfer time helps reduce the electric loss. If the system is well balanced and friction is neglected, the electric loss is proportional to $\int_0^T a^2 dt$ which is then defined as a measure of electric loss. Integrate (44) using (7) then substitute (8) into the result

$$E_{loss} \propto \int_0^T a^2 dt = \frac{2}{3} A^2 (2T_j + 3T_A) \quad (45)$$

By using Table I, (45) can be calculated for 4 types of S-curve:

$$E_{loss} \propto \begin{cases} \frac{2}{3} J_{max} L, & 3 \text{ segments} \\ \left(\sqrt{\left(\frac{A_{max}^3}{J_{max}} \right)^2 + 4A_{max}^3 L} - \frac{5}{3} \frac{A_{max}^3}{J_{max}} \right), & 5 \text{ segments A} \\ \frac{4}{3} \sqrt{V_{max} J_{max}}, & 5 \text{ segments B} \\ \left(2V_{max} A_{max} - \frac{2}{3} \frac{A_{max}^3}{J_{max}} \right), & 7 \text{ segments} \end{cases} \quad (46)$$

For $R \leq 1$, the time-optimal S-curve profile is 3-segment for $L \leq L_2$ and 5-segment type-B for $L > L_2$, as in Fig. 10.

For $R > 1$, the time-optimal S-curve profile is 3-segment for $L \leq L_1$, 5-segment type-A for $L_1 < L \leq L_3$, and 7-segment for $L > L_3$, as in Fig. 11.

Equation (46) shows that the electric loss remains unchanged as motion length increases when the time-optimal S-curve is 5-segment type-B or 7-segment. This is because in these cases, increasing motion length leads to an increase in the constant-velocity time T_v which corresponds to $a = 0$.

For fixed values of V_{max} , A_{max} , increasing J_{max} leads to an increase of R . Equation (46) shows that the electric loss increases as jerk limit increases, as in Fig. 12.

IV. CONCLUSION

In conclusion, the time optimization of S-curve profile has been solved for all ranges of motion length and limits by investigating four types of its shape. The transfer time and electric loss during the point-to-point motion of the system using S-curve profile have also been analyzed for all cases. This is applicable to the control of point-to-point motion in robots, CNC machines, wire boding machines, etc.

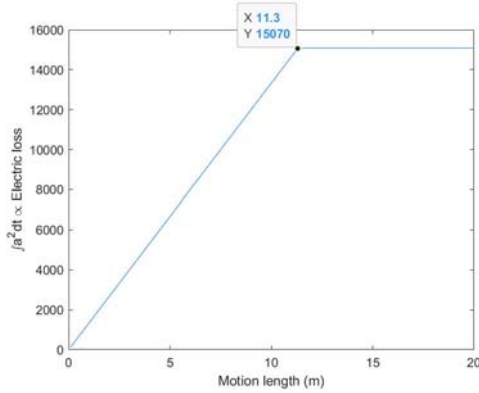


Fig. 10. Measure of electric loss for different motion lengths, given $V_{max} = 40ms^{-1}$, $A_{max} = 400ms^{-2}$, $J_{max} = 2000ms^{-3}$, hence $R = 0.5$ and $L_2 = 11.3m$.

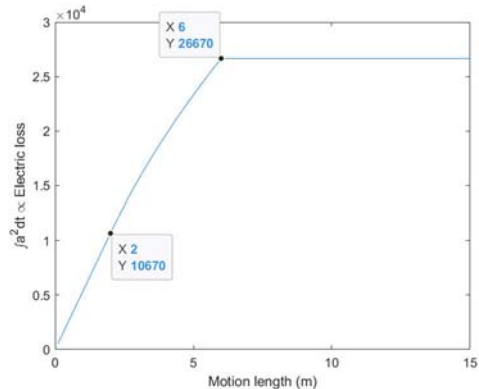


Fig. 11. Measure of electric loss for different motion lengths, given $V_{max} = 40ms^{-1}$, $A_{max} = 400ms^{-2}$, $J_{max} = 8000ms^{-3}$, hence $R = 2$, $L_1 = 2m$, and $L_3 = 6m$.

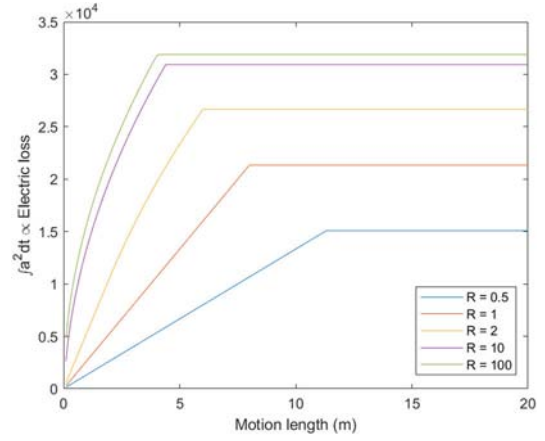


Fig. 12. Measure of electric loss for different motion lengths, $R = 0.5, 1, 2, 10, 100$ by increasing J_{max} , given $V_{max} = 40ms^{-1}$, $A_{max} = 400ms^{-2}$.

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